D · prototype NNs · round 1

proto	trained	eval	nNN1	nNN2	AUC1	min1	AUC2	min2	dCor	rho
P0	endcap	endcap	16	10	0.880	0.870	0.984	0.962	0.146	+0.181
P0	barrel	barrel	20	10	0.903	0.837	0.950	0.894	0.130	+0.092
P1	endcap	endcap	9	8	0.861	0.852	0.983	0.963	0.177	+0.228
P1	barrel	barrel	11	6	0.898	0.838	0.947	0.888	0.144	+0.105
P2	endcap	endcap	9	14	0.861	0.852	0.987	0.971	0.148	+0.179
P2	barrel	barrel	14	9	0.992	0.988	0.987	0.982	0.144	+0.088
P3	inclusive	endcap	15	10	0.930	0.925	0.983	0.963	0.109	+0.160
P3	inclusive	barrel	15	10	0.919	0.848	0.950	0.882	0.126	+0.142

AUC = cross-validated (5-fold, out-of-fold) AUC of a gradient-boosted surrogate,
signal mass points weighted equally; min = worst single mass point vs background.
dCor / rho = dependence between the NN1 and NN2 surrogate scores on background (0 = independent).
Barrel-only vs inclusive: compare P1 trained=barrel with P3 trained=inclusive, eval=barrel.
The surrogate nominates; it does not crown. Final placement is GAMBIT's (closure and G).

D · P0 · mk4 as is (1/2)

P0: mk4 as is

NN1 endcap (16)

nMSeg_ratio_EIEM	D_endcap	0.086	block 60
track_scalar_sum_pt_endcap	D_endcap	0.286	block 34
msvtx_caloClusterSumE	D_endcap	0.383	block 16
MS1Vtx_avg_clusE_raw	D_endcap	0.361	block 22
MS1Vtx_rms_clusE_raw	D_endcap	0.365	block 22
MS1Vtx_maxclusE_raw	D_endcap	0.374	block 16
msvtx_el1ecal	D_endcap	0.324	block 28
msvtx_el1hcal	D_endcap	0.313	block 29
msvtx_el2ecal	D_endcap	0.343	block 23
msvtx_el2hcal	D_endcap	0.311	block 31
msvtx_el3ecal	D_endcap	0.364	block 23
msvtx_el3hcal	D_endcap	0.193	block 44
msvtx_el4ecal	D_endcap	0.337	block 25
msvtx_el4hcal	D_endcap	0.094	block 58
MS1Vtx_mindR_jetcut_raw	D_endcap	0.308	block 32
msvtx_trackSoftSumPt_NOSYS	D_endcap	0.140	block 47

NN2 endcap (10)

MSTracklet_endcap_avg_dR	D_endcap	0.399	block 15
MSTracklet_endcap_rms_dR	D_endcap	0.399	block 15
MSeg_endcap_avg_dR	D_endcap	0.462	block 13
MSeg_endcap_rms_dR	D_endcap	0.384	block 13
MS1Vtx_endcap_hits_ntot	D_endcap	0.718	block 4
MSVtx_MET_dphi_raw	D_endcap	0.375	block 19
MS1Vtx_maxclustime_raw	D_endcap	0.356	block 24
MS1Vtx_avg_clustime_raw	D_endcap	0.229	block 40
MS1Vtx_rms_clustime_raw	D_endcap	0.335	block 26
msvtx_nTrk	D_endcap	0.606	block 9

NN1 barrel (20)

nMSeg_ratio_BIBM	D_barrel	0.263	block 31
track_scalar_sum_pt_barrel	D_barrel	0.265	block 30
msvtx_caloClusterSumE	D_barrel	0.310	block 24
MS1Vtx_avg_clusE_raw	D_barrel	0.304	block 24
MS1Vtx_rms_clusE_raw	D_barrel	0.299	block 24
MS1Vtx_maxclusE_raw	D_barrel	0.312	block 24
msvtx_el1ecal	D_barrel	0.235	block 37
msvtx_el1hcal	D_barrel	nan	block -
msvtx_el2ecal	D_barrel	0.267	block 25
msvtx_el2hcal	D_barrel	0.259	block 33
msvtx_el3ecal	D_barrel	0.293	block 25
msvtx_el3hcal	D_barrel	0.240	block 35
msvtx_el4ecal	D_barrel	0.243	block 34
msvtx_el4hcal	D_barrel	0.166	block 43
MS1Vtx_mindR_jetcut_raw	D_barrel	0.365	block 18
msvtx_trackSoftSumPt_NOSYS	D_barrel	0.111	block 53
msvtx_caloClusterSumPtScalar	D_barrel	0.307	block 24
msegUnAssoc_counts/nBIL	D_barrel	0.152	block 47
msegUnAssoc_counts/nBML	D_barrel	0.277	block 26
nBOL_NOSYS	D_barrel	0.319	block 19

NN2 barrel (10)

MSTracklet_barrel_avg_dR	D_barrel	0.097	block 56
MSTracklet_barrel_rms_dR	D_barrel	0.097	block 56
MSeg_barrel_avg_dR	D_barrel	0.366	block 17
MSeg_barrel_rms_dR	D_barrel	0.297	block 17
MS1Vtx_barrel_hits_ntot	D_barrel	0.632	block 9

D · P0 · mk4 as is (2/2)

```
   MSVtx_MET_dphi_raw                     D_barrel  0.429   block 14
   MS1Vtx_maxclustime_raw                 D_barrel  0.274   block 29
   MS1Vtx_avg_clustime_raw                D_barrel  0.157   block 45
   MS1Vtx_rms_clustime_raw                D_barrel  0.262   block 32
   msvtx_nTrk                             D_barrel  0.442   block 12

changes:
  mk4 as it stands (list names resolved to scored columns; duplicates collapsed)
```

D · P1 · mk4 minus weak/dead/redundant (1/2)

```
P1: mk4 minus weak/dead/redundant

NN1 endcap  (9)
   msvtx_caloClusterSumE                  D_endcap  0.383   block 16
   MS1Vtx_rms_clusE_raw                   D_endcap  0.365   block 22
   msvtx_el3ecal                          D_endcap  0.364   block 23
   msvtx_el4ecal                          D_endcap  0.337   block 25
   msvtx_el1ecal                          D_endcap  0.324   block 28
   msvtx_el1hcal                          D_endcap  0.313   block 29
   msvtx_el2hcal                          D_endcap  0.311   block 31
   MS1Vtx_mindR_jetcut_raw                D_endcap  0.308   block 32
   track_scalar_sum_pt_endcap             D_endcap  0.286   block 34

NN2 endcap  (8)
   MS1Vtx_endcap_hits_ntot                D_endcap  0.718   block 4
   msvtx_nTrk                             D_endcap  0.606   block 9
   MSeg_endcap_avg_dR                     D_endcap  0.462   block 13
   MSTracklet_endcap_avg_dR               D_endcap  0.399   block 15
   MSVtx_MET_dphi_raw                     D_endcap  0.375   block 19
   MS1Vtx_maxclustime_raw                 D_endcap  0.356   block 24
   MS1Vtx_rms_clustime_raw                D_endcap  0.335   block 26
   MS1Vtx_avg_clustime_raw                D_endcap  0.229   block 40

NN1 barrel  (11)
   MS1Vtx_mindR_jetcut_raw                D_barrel  0.365   block 18
   nBOL_NOSYS                             D_barrel  0.319   block 19
   MS1Vtx_maxclusE_raw                    D_barrel  0.312   block 24
   msvtx_el3ecal                          D_barrel  0.293   block 25
   msegUnAssoc_counts/nBML                D_barrel  0.277   block 26
   track_scalar_sum_pt_barrel             D_barrel  0.265   block 30
   nMSeg_ratio_BIBM                       D_barrel  0.263   block 31
   msvtx_el2hcal                          D_barrel  0.259   block 33
   msvtx_el4ecal                          D_barrel  0.243   block 34
   msvtx_el3hcal                          D_barrel  0.240   block 35
   msvtx_el1ecal                          D_barrel  0.235   block 37

NN2 barrel  (6)
   MS1Vtx_barrel_hits_ntot                D_barrel  0.632   block 9
   msvtx_nTrk                             D_barrel  0.442   block 12
   MSVtx_MET_dphi_raw                     D_barrel  0.429   block 14
   MSeg_barrel_avg_dR                     D_barrel  0.366   block 17
   MS1Vtx_maxclustime_raw                 D_barrel  0.274   block 29
   MS1Vtx_rms_clustime_raw                D_barrel  0.262   block 32

changes:
  NN1 endcap: drop MS1Vtx_maxclusE_raw (same block as msvtx_caloClusterSumE, |rho| >= 0.9)
  NN1 endcap: drop MS1Vtx_avg_clusE_raw (same block as MS1Vtx_rms_clusE_raw, |rho| >= 0.9)
  NN1 endcap: drop msvtx_el2ecal (same block as msvtx_el3ecal, |rho| >= 0.9)
  NN1 endcap: drop msvtx_el3hcal (D 0.193, weak or dead)
  NN1 endcap: drop msvtx_trackSoftSumPt_NOSYS (D 0.140, weak or dead)
  NN1 endcap: drop msvtx_el4hcal (D 0.094, weak or dead)
  NN1 endcap: drop nMSeg_ratio_EIEM (D 0.086, weak or dead)
  NN2 endcap: drop MSTracklet_endcap_rms_dR (same block as MSTracklet_endcap_avg_dR, |rho| >= 0.9)
  NN2 endcap: drop MSeg_endcap_rms_dR (same block as MSeg_endcap_avg_dR, |rho| >= 0.9)
  NN1 barrel: drop msvtx_caloClusterSumE (same block as MS1Vtx_maxclusE_raw, |rho| >= 0.9)
  NN1 barrel: drop msvtx_caloClusterSumPtScalar (same block as MS1Vtx_maxclusE_raw, |rho| >= 0.9)
  NN1 barrel: drop MS1Vtx_avg_clusE_raw (same block as MS1Vtx_maxclusE_raw, |rho| >= 0.9)
  NN1 barrel: drop MS1Vtx_rms_clusE_raw (same block as MS1Vtx_maxclusE_raw, |rho| >= 0.9)
  NN1 barrel: drop msvtx_el2ecal (same block as msvtx_el3ecal, |rho| >= 0.9)
  NN1 barrel: drop msvtx_el4hcal (D 0.166, weak or dead)
```

D · P1 · mk4 minus weak/dead/redundant (2/2)

```
NN1 barrel: drop msegUnAssoc_counts/nBIL (D 0.152, weak or dead)
NN1 barrel: drop msvtx_trackSoftSumPt_NOSYS (D 0.111, weak or dead)
NN1 barrel: drop msvtx_el1hcal (D nan, weak or dead)
NN2 barrel: drop MSeg_barrel_rms_dR (same block as MSeg_barrel_avg_dR, |rho| >= 0.9)
NN2 barrel: drop MS1Vtx_avg_clustime_raw (D 0.157, weak or dead)
NN2 barrel: drop MSTracklet_barrel_avg_dR (D 0.097, weak or dead)
NN2 barrel: drop MSTracklet_barrel_rms_dR (D 0.097, weak or dead)
```

D · P2 · P1 + strong unused (non-HTmiss) features (1/2)

```
P2: P1 + strong unused (non-HTmiss) features

NN1 endcap  (9)
   msvtx_caloClusterSumE                  D_endcap  0.383   block 16
   MS1Vtx_rms_clusE_raw                   D_endcap  0.365   block 22
   msvtx_el3ecal                          D_endcap  0.364   block 23
   msvtx_el4ecal                          D_endcap  0.337   block 25
   msvtx_el1ecal                          D_endcap  0.324   block 28
   msvtx_el1hcal                          D_endcap  0.313   block 29
   msvtx_el2hcal                          D_endcap  0.311   block 31
   MS1Vtx_mindR_jetcut_raw                D_endcap  0.308   block 32
   track_scalar_sum_pt_endcap             D_endcap  0.286   block 34

NN2 endcap  (14)
   MS1Vtx_endcap_hits_ntot                D_endcap  0.718   block 4
   msvtx_nTrk                             D_endcap  0.606   block 9
   MSeg_endcap_avg_dR                     D_endcap  0.462   block 13
   MSTracklet_endcap_avg_dR               D_endcap  0.399   block 15
   MSVtx_MET_dphi_raw                     D_endcap  0.375   block 19
   MS1Vtx_maxclustime_raw                 D_endcap  0.356   block 24
   MS1Vtx_rms_clustime_raw                D_endcap  0.335   block 26
   MS1Vtx_avg_clustime_raw                D_endcap  0.229   block 40
   msvtx_nTGCOuterStation                 D_endcap  0.725   block 3
   msvtx_nTGC                             D_endcap  0.701   block 5
   msvtx_nTGCExtendedStation              D_endcap  0.700   block 6
   msvtx_nMDTMiddleStation                D_endcap  0.648   block 8
   msvtx_nTGCMiddleStation                D_endcap  0.601   block 10
   msvtx_nMDTOuterStation                 D_endcap  0.583   block 11

NN1 barrel  (14)
   MS1Vtx_mindR_jetcut_raw                D_barrel  0.365   block 18
   nBOL_NOSYS                             D_barrel  0.319   block 19
   MS1Vtx_maxclusE_raw                    D_barrel  0.312   block 24
   msvtx_el3ecal                          D_barrel  0.293   block 25
   msegUnAssoc_counts/nBML                D_barrel  0.277   block 26
   track_scalar_sum_pt_barrel             D_barrel  0.265   block 30
   nMSeg_ratio_BIBM                       D_barrel  0.263   block 31
   msvtx_el2hcal                          D_barrel  0.259   block 33
   msvtx_el4ecal                          D_barrel  0.243   block 34
   msvtx_el3hcal                          D_barrel  0.240   block 35
   msvtx_el1ecal                          D_barrel  0.235   block 37
   msegUnAssoc_counts/nEML                D_barrel  0.858   block 3
   nMSeg_ratio_EIEM                       D_barrel  0.755   block 5
   nEOL_NOSYS                             D_barrel  0.710   block 8

NN2 barrel  (9)
   MS1Vtx_barrel_hits_ntot                D_barrel  0.632   block 9
   msvtx_nTrk                             D_barrel  0.442   block 12
   MSVtx_MET_dphi_raw                     D_barrel  0.429   block 14
   MSeg_barrel_avg_dR                     D_barrel  0.366   block 17
   MS1Vtx_maxclustime_raw                 D_barrel  0.274   block 29
   MS1Vtx_rms_clustime_raw                D_barrel  0.262   block 32
   msvtx_nRPCOuterStation                 D_barrel  0.815   block 4
   msvtx_nRPC                             D_barrel  0.725   block 7
   msvtx_nRPCMiddleStation                D_barrel  0.520   block 11

changes:
  endcap: add msvtx_nTGCOuterStation to NN2 (D 0.725; max |rho| NN1 0.04, NN2 0.73 -> joins the net it already correlates with)
  endcap: add msvtx_nTGC to NN2 (D 0.701; max |rho| NN1 0.06, NN2 0.81 -> joins the net it already correlates with)
  endcap: add msvtx_nTGCExtendedStation to NN2 (D 0.700; max |rho| NN1 0.05, NN2 0.66 -> joins the net it already correlates with)
```

D · P2 · P1 + strong unused (non-HTmiss) features (2/2)

```
endcap: add msvtx_nMDTMiddleStation to NN2 (D 0.648; max |rho| NN1 0.05, NN2 0.76 -> joins the net it already correlates with)
endcap: add msvtx_nTGCMiddleStation to NN2 (D 0.601; max |rho| NN1 0.14, NN2 0.78 -> joins the net it already correlates with)
endcap: add msvtx_nMDTOuterStation to NN2 (D 0.583; max |rho| NN1 0.07, NN2 0.73 -> joins the net it already correlates with)
barrel: add msegUnAssoc_counts/nEML to NN1 (D 0.858; max |rho| NN1 0.11, NN2 0.07 -> joins the net it already correlates with)
barrel: add msvtx_nRPCOuterStation to NN2 (D 0.815; max |rho| NN1 0.32, NN2 0.54 -> joins the net it already correlates with)
barrel: add nMSeg_ratio_EIEM to NN1 (D 0.755; max |rho| NN1 0.50, NN2 0.05 -> joins the net it already correlates with)
barrel: add msvtx_nRPC to NN2 (D 0.725; max |rho| NN1 0.32, NN2 0.87 -> joins the net it already correlates with)
barrel: add nEOL_NOSYS to NN1 (D 0.710; max |rho| NN1 0.49, NN2 0.06 -> joins the net it already correlates with)
barrel: add msvtx_nRPCMiddleStation to NN2 (D 0.520; max |rho| NN1 0.20, NN2 0.78 -> joins the net it already correlates with)
```

D · P3 · inclusive P1 with region flag

```
P3: inclusive P1 with region flag

NN1 inclusive  (15)
   msvtx_caloClusterSumE                  D_endcap  0.383   block 16
   MS1Vtx_rms_clusE_raw                   D_endcap  0.365   block 22
   msvtx_el3ecal                          D_endcap  0.364   block 23
   msvtx_el4ecal                          D_endcap  0.337   block 25
   msvtx_el1ecal                          D_endcap  0.324   block 28
   msvtx_el1hcal                          D_endcap  0.313   block 29
   msvtx_el2hcal                          D_endcap  0.311   block 31
   MS1Vtx_mindR_jetcut_raw                D_endcap  0.308   block 32
   track_scalar_sum_pt_endcap             D_endcap  0.286   block 34
   nBOL_NOSYS                             D_endcap  0.271   block 35
   MS1Vtx_maxclusE_raw                    D_endcap  0.374   block 16
   msegUnAssoc_counts/nBML                D_endcap  0.069   block 61
   track_scalar_sum_pt_barrel             D_endcap  0.213   block 42
   nMSeg_ratio_BIBM                       D_endcap  0.330   block 27
   msvtx_el3hcal                          D_endcap  0.193   block 44

NN2 inclusive  (10)
   MS1Vtx_endcap_hits_ntot                D_endcap  0.718   block 4
   msvtx_nTrk                             D_endcap  0.606   block 9
   MSeg_endcap_avg_dR                     D_endcap  0.462   block 13
   MSTracklet_endcap_avg_dR               D_endcap  0.399   block 15
   MSVtx_MET_dphi_raw                     D_endcap  0.375   block 19
   MS1Vtx_maxclustime_raw                 D_endcap  0.356   block 24
   MS1Vtx_rms_clustime_raw                D_endcap  0.335   block 26
   MS1Vtx_avg_clustime_raw                D_endcap  0.229   block 40
   MS1Vtx_barrel_hits_ntot                D_endcap    nan   block -
   MSeg_barrel_avg_dR                     D_endcap    nan   block -

changes:
  inclusive: one net per role trained on BOTH regions with a region flag input;
  features = union of P1 endcap and P1 barrel (region-tagged columns are missing outside
  their region; the surrogate handles missing values natively). Scored per region.
```